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Project Title:	Controlling NAD(P) Hyper-Oxidation to Regulate Repair and Maintenance Processes in Humans in Space		
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Comments:			
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COI Name (Institution):	Gassman, Natalie Ph.D. (University of Alabama at Birmingham) Nickels, Bryce Ph.D. (Rutgers University) Pluth, Janice Ph.D. (University of Nevada, Las Vegas)		
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Task Description:	Exposure to ionizing radiation, microgravity, and metabolic stressors during spaceflight causes excessive reactive oxygen species (ROS), contributing to cellular stress and damage in astronauts. The stresses of space travel alter genetic and epigenetic makeup, metabolism, mitochondrial function, transcription, and translation in cells. These changes affect the health and performance of astronauts in space and long after their return. Oxidized macromolecules are pivotal actors in the loss of cell function and integrity. In space, strategies using anti-oxidants to reduce ROS have been ineffectual because they do not target the pathogenic species that most heighten dysfunction. Macromolecules involved in cellular maintenance and repair catalyze biochemical processes that use cofactors, mostly derived from dietary B-vitamins. Since vitamins and cofactors can be subject to ROS-accelerated degradation, compounded by food processing, their availability is reduced during space travel. One such limited species is Nicotinamide Adenine Dinucleotide (Phosphate), or NAD(P), a cofactor derived from vitamin B3 that enables biochemical processes directed to combatting ROS and maintaining cellular homeostasis. NAD(P) shortage disrupts bioenergetic processes, central metabolism, chromatin integrity, hormone functionalization, drug metabolism, and ROS regulation. Crucially, NAD(P) is hyper-oxidized by ROS to non-redox entities called ox-NAD(P). Ox-NAD(P) species are so similar to NAD(P) that they directly interfere with the same enzymatic processes, becoming potentiators of dysfunction. We believe that in space, ox-NAD(P) are critical, ever-increasing, pathogenic entities disrupting repair and maintenance processes necessary to astronauts' health and performance, and become more of an issue as dietary sources of NAD(P) decline. We propose that intracellular ox-NAD(P) and derivatives are more abundant under ROS-promoting spaceflight stressor conditions, and cause more acute cell-wide dysfunction in space than when on Earth. We hypothesize that tissues with high demand for NAD(P), such as liver, heart, blood, muscle, and skin, will be more adversely affected by ox-NAD(P) upon NAD shortage than other tissues, positioning ox-NAD(P) as a major protagonist in the accelerated loss of endogenous repair and maintenance processes in space. Critically, in studying factors which improved metabolic function we noted a corresponding decrease in ox-NAD(P), suggesting a health benefit to reducing ox-NAD(P). To mitigate the effects of inflight ROS-generated ox-NAD(P) on cell homeostasis, we will establish the full effects of the synergy of space-stressors on NAD(P) hyperoxidation following exposure to each stressor individually and concurrently; characterize ox-NAD(P)’s effects on cellular metabolism, DNA repair, and gene expression; and finally, identify a unique intervention that offers the most effective means to control ox-NAD(P) production and promote optimal cellular function. Applying novel chemistry, enzymology, and metabolomics, we will evaluate changes in cell bioenergetics, DNA double-strand breaks (DSB), and cell growth, and measure effects on transcription and translation using human cell lines. Finally, we will examine the effectiveness of the remediation achieved through reduced levels of ox-NAD(P) under space stressor conditions with the intent to improve astronauts' health and well-being. This study looks, for the first time, at the pivotal role of ox-NAD(P) in cellular dysfunction and its abundance as an indicator of cellular impairment, and proposes countermeasures that mitigate its formation to aid in restoring optimal cellular homeostasis to reduce risks to astronauts' health and performance during deep space explorations. This project will reveal biomarkers of cellular decline, inhibitors of repair and cellular maintenance, and offer a controllable and reversible approach to lessen their abundance and negate their adverse activity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. Original Projects Aims - Background: Exposure to ionizing radiation, microgravity, and metabolic stressors during spaceflight causes excessive reactive oxygen species (ROS) contribute to cellular stress and damage. Biochemical cellular maintenance and repair processes mostly use cofactors derived from dietary B-vitamins. One such species is Nicotinamide Adenine Dinucleotide (Phosphate), NAD(P), a cofactor derived from vitamin B3. NAD(P) enables the biochemical processes that maintain cellular homeostasis and combat ROS. NAD(P) shortage disrupts bioenergetic processes, central metabolism, chromatin integrity, and ROS regulation. NAD(P) is also hyper-oxidized by ROS to non-redox entities called ox-NAD(P), species that compete against NAD(P) in catalysis. These entities are present in cells, tissues, and human biospecimens at physiologically relevant concentrations under ROS conditions. These ox-NAD(P) species could be a missing link between metabolic imbalance, ROS, and loss of cellular homeostasis. Considering the breadth of bioenergetic, biosynthetic, and signaling processes that are under NAD(P) control, risks associated with the accumulation of ox-NAD(P) and the reduction of NAD levels we anticipate the negative effects of these entities to be more pronounced with ROS-inducing spaceflight stressors. Hypothesis: Space-flight stressors promote ox-NAD(P) formation by reaction of NAD(P) with superoxide and other ROS-related species. Ox-NAD(P) accumulation disrupts cellular bioenergetics, macromolecule biosynthesis and alters DNA repair processes, processes that are amplified by declining NAD levels. Restoration of NAD levels will slow down the nefarious effects of ox-NAD(P) accumulation. Significance: This study looks at the pivotal role of endogenous ox-NAD(P) in cellular dysfunction and its abundance as an indicator of cellular impairment and proposes countermeasures that mitigate its formation to aid in restoring optimal cellular homeostasis to reduce risks to astronauts’ health and performance during deep space explorations. Objectives: The project seeks to reveal a fundamental and multi modal cause of dysfunction in endogenous repair and cellular maintenance processes and provide the means to mitigate it. The specific aims are as follows: Specific Aim 1: Define the impact of mimicking spaceflight-induced stress on ox-NAD(P) levels and their derivatives in human cells in a 2D and a 3D tissue context. Specific Aim 2: Define the impact of alterations in ox-NAD(P) levels on metabolism, ROS, proteostasis (transcription and translation), and DNA damage in human cells. Specific Aim 3: Evaluate the reversible and protective effect of reducing ox-NAD(P) levels with nicotinamide riboside in human cells under space stressor conditions and in a clinical trial. 2. Key Findings: - We completed the synthesis of the 2, 4, and 6- isomers of ox-NAD, and 2, and 6-isomer of ox-NADP and the synthesis of isotopolog M+2 PYR and 4-ox-NAD. The synthesis of the PYR-MP and TP series is also underway but the purification of the PYR-triphosphates remains challenging. - The synthesis of novel membrane-permeable ox-NAD(P) derivatives has been initiated and we have managed to generate small quantities of the masked 4-ox-NAD and demonstrated on HepG3 cell cultures that this modification led to cell death while the non-masked extracellular ox-NAD remained inactive. We are therefore all set to apply this

	chemistry on all ox-NAD(P) and PYR-TP species in year 2. - We have initiated chemistry to generate ox-NAD(P) and PYR-TP that incorporate conjugatable appendages. We have generated the pyridone riboside for each isomers that now incorporate an ethyl-azido moiety on the amide functionality of PY and used these modified nucleobases to generate the nucleoside derivatives. These will be used to establish RNA interference in the coming months. As soon as the triphosphates becomes available they will be used in vitro to examine RNA incorporation. - Metabolomics of cell extracts following exposure to acetaminophen have revealed an increase in ox-NAD levels, and changes in NAD/NADH ratios and NAD/NADH ratios. We observed the possible presence of two new NAD metabolites, being synthesized to confirm their identity. These could be interfering with DNA replication and repair as they accumulate. This presents an unexpected project development. - In preliminary metabolic experiments on HepG3 cells, we observed that 4-PYR affects OCR and ECAR by seahorse. In a mitofuel assay, an effect on the glutamine and glucose consumption was also observed with cells preferring using lipids when 4-PYR was present. Additional discoveries: Ox-NAD are inhibitors of the commercially available NADGlo cycling assay that employ luciferase and an NAD dependent oxidoreductase. We observed that each isomer inhibited these two enzymes used in the kit to different degree. Importantly, biospecimens that contain higher levels of ox-NAD might provide measurements of NAD(H) that are below the actual levels, due to this inhibition. Ox-NAD also appears to be a modulator of PARP-1 autoMARylation activity, although at high ox-NAD concentration. Potentially, this observation has consequences on DNA damage repair processes. Also, ox-NAD increases the non-productive consumption of NAD by increasing NADase activity in PARP-1. 3. Impacts of Key Finding on hypotheses, technology requirements, objectives and specific aims of the original proposal. The hypothesis, objectives, and specific aims remain the same. We have introduced new chemical synthesis to our workflow to better characterize the newly identified 2'-deoxy-PYR series. These key findings reinforce our interest in understanding the role of ox-NAD(P) in cellular dysfunction and explore how its abundance can be reduced. 4. Proposed research plan for the coming year: The continuation and completion of the proposed research plan remains unchanged.
Bibliography Type:	Description: (Last Updated: 11/07/2025)
Articles in Peer-reviewed Journals	Hayat F, Deason JT, Bryan RL, Terkeltaub R, Song W, Kraus WL, Pluth J, Gassman NR, Migaud ME. "Synthesis, detection, and metabolism of pyridone ribosides, products of NAD overoxidation." Chem Res Toxicol. 2024 Feb 19;37(2):248-58. https://doi.org/10.1021/acs.chemrestox.3c00264 ; PMID: 38198686; PMCID: PMC10880730 , Feb-2024
Awards	Migaud M. "Research & Technology Showcase Award, Alabama, USA. " Oct-2023